

NEWPORT, VT, USA

WMO: 720493

Lat: 44.883N

Lon: 72.233W

Elev: 279

StdP: 98.02

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 54758

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.0	-21.1	-28.5	0.3	-22.9	-26.0	0.4	-19.8	10.4	-2.5	9.1	-4.5	1.6	220	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	29.0	20.7	27.5	19.1	26.1	18.9	22.2	26.5	21.3	25.3	20.3	24.2	3.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	16.0	24.4	19.9	15.1	23.6	18.9	14.1	22.6	66.5	26.7	63.0	25.3	59.4	24.2	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.4	6.5	-28.3	31.6	2.7	1.2	-30.3	32.5	-31.9	33.2	-33.4	33.8	-35.4	34.7		
(5)			-28.6	23.6	2.5	1.1	-30.4	24.4	-31.8	25.0	-33.2	25.6	-35.0	26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-8.6	-6.8	-2.7	5.1	12.3	16.3	19.6	18.6	14.8	8.2	1.5	-4.5		
	DBStd	11.07	6.94	6.41	6.33	5.03	4.52	3.71	3.02	2.85	4.25	4.72	5.46	6.44		
	HDD10.0	2447	577	472	396	164	27	2	0	0	7	93	259	450		
	HDD18.3	4572	835	705	652	398	194	83	22	32	122	315	506	708		
	CDD10.0	1059	0	0	2	17	98	190	297	267	149	37	3	0		
	CDD18.3	144	0	0	0	1	7	22	60	40	15	0	0	0		
	CDH23.3	1143	0	0	2	10	70	191	477	273	117	3	0	0		
	CDH26.7	219	0	0	0	2	9	41	105	41	21	0	0	0		
(14) Wind	WSAvg	2.6	2.7	2.8	3.1	3.2	2.7	2.3	2.1	2.0	2.1	2.6	2.7	2.5		
(15) Precipitation	PrecAvg	1015	62	58	69	75	89	104	101	111	91	81	91	82		
	PrecMax	1333	166	111	109	144	169	273	184	190	197	179	167	181		
	PrecMin	777	13	11	17	22	30	43	39	46	27	19	47	40		
	PrecStd	151	35	25	24	27	40	53	37	39	44	34	28	37		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.9	10.2	22.8	24.8	27.9	30.7	31.3	29.9	29.1	22.7	17.6	10.2		
		MCWB	6.8	7.5	14.5	14.8	18.9	21.0	21.2	20.9	22.1	16.2	12.0	7.9		
	2%	DB	5.4	6.3	12.0	19.9	25.2	27.7	29.4	27.6	26.4	19.9	14.4	7.7		
		MCWB	3.4	3.5	7.1	12.7	16.6	18.6	21.1	19.3	19.4	14.2	10.4	6.1		
	5%	DB	2.9	4.1	8.0	17.0	22.8	25.8	27.6	26.2	24.5	17.6	12.2	5.8		
		MCWB	1.2	1.6	4.4	10.8	15.5	17.8	19.5	19.3	18.7	13.3	8.7	4.2		
	10%	DB	1.0	2.3	5.9	13.1	20.9	23.3	26.1	24.9	22.4	15.8	9.6	3.2		
		MCWB	-0.5	0.5	2.5	7.9	14.2	17.1	19.3	18.9	17.6	12.4	6.9	1.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	8.6	14.2	16.2	20.4	22.3	24.1	22.8	22.5	18.4	13.4	8.3		
		MCDB	7.9	9.1	22.5	21.6	26.1	27.3	29.2	27.0	27.0	20.1	15.3	9.6		
	2%	WB	4.0	4.2	7.5	14.1	18.7	20.8	22.6	21.5	21.0	15.8	11.3	6.6		
		MCDB	5.3	5.9	10.8	19.0	23.4	25.2	26.6	25.4	24.9	17.9	13.8	7.7		
	5%	WB	1.5	2.0	4.9	11.2	16.9	19.5	21.6	20.5	19.8	14.3	9.0	4.4		
		MCDB	3.0	3.9	7.8	15.0	21.0	24.0	25.5	24.0	23.6	17.6	11.4	5.8		
	10%	WB	-0.2	0.4	2.9	8.7	15.3	18.2	20.4	19.7	18.4	12.6	6.8	2.1		
		MCDB	0.8	1.8	5.5	13.4	19.4	21.8	24.4	23.4	21.5	15.2	9.3	3.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.2	9.3	9.8	11.3	10.5	10.7	10.5	10.9	8.9	7.6	7.3		
		MCDBR	8.5	9.9	11.3	15.3	13.9	13.6	12.6	12.2	13.0	11.6	11.7	8.7		
		MCWBR	8.2	8.1	7.6	9.1	7.6	6.4	6.0	6.1	7.2	7.5	8.4	8.0		
		MCDBR	8.4	9.7	10.3	13.3	11.6	10.8	10.7	10.4	11.9	10.3	10.4	8.2		
	5% WB	MCWBR	8.2	8.5	7.6	8.8	7.3	5.9	5.9	5.9	7.3	7.6	8.7	8.0		
(40) Clear-Sky Solar Irradiance	taub		0.258	0.281	0.317	0.349	0.367	0.416	0.425	0.398	0.361	0.324	0.304	0.272		
	taud		2.389	2.299	2.334	2.385	2.396	2.249	2.238	2.304	2.406	2.514	2.493	2.408		
	Ebn at Noon		873	907	917	912	902	854	841	852	859	852	810	813		
	Edh at Noon		73	97	108	113	115	134	134	121	101	79	67	65		
(44) All-Sky Solar Radiation	RadAvg		1.41	2.30	3.46	4.27	4.97	5.45	5.51	4.94	3.97	2.31	1.42	1.02		
	RadStd		0.17	0.26	0.24	0.51	0.54	0.46	0.43	0.26	0.31	0.23	0.19	0.09		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+65	N/A

Nomenclature: See separate page